



## Lab 05: Routing Between Different Networks

Router0 (2911) · Switch0 · Switch1 · PC0 · Laptop0 · PC1

Lab #05

Topic: Inter-Network Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

### 1. Lab Objectives

- Connect two 2960 switches to a single 2911 router on separate Gig interfaces.
- Place PC0 and Laptop0 on Switch0, and PC1 on Switch1.
- Router0 provides routing between the two switch subnets.
- Verify cross-subnet connectivity from PC0/Laptop0 to PC1.
- Relate to Network+ objective 1.4: Routing concepts and path determination.

### 2. Network Topology

Router0 (2911) is at the top. Gig0/0 links to Switch0 (2960-24TT) serving PC0 (Fa0/1) and Laptop0 (Fa0/3). Gig0/1 links to Switch1 (2960-24TT) serving PC1 (Fa0/1). Two subnets: 192.168.1.0/24 left side, 192.168.2.0/24 right side.

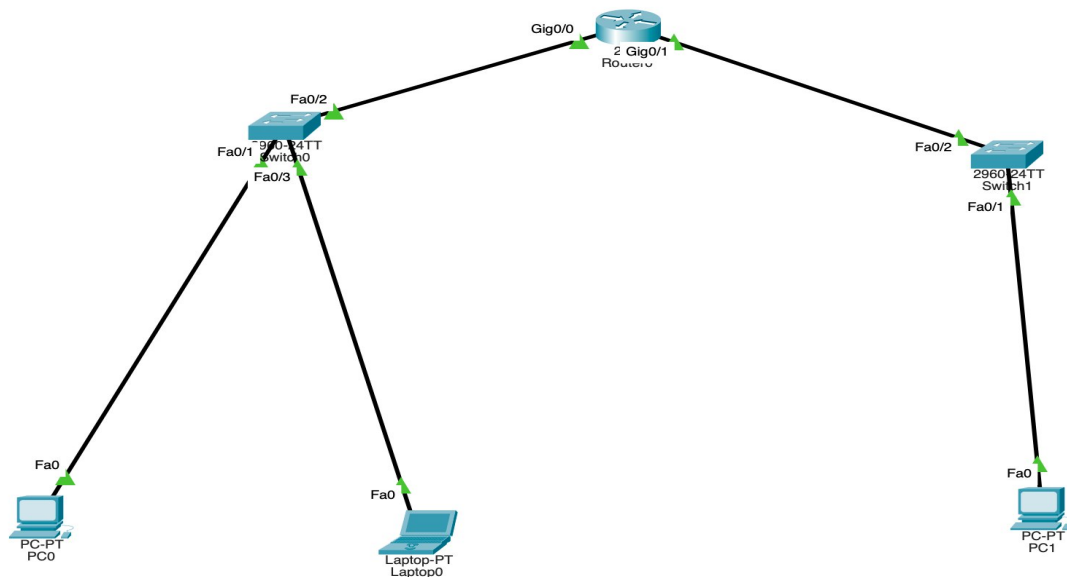


Figure 1 – Router0 connecting Switch0 (PC0, Laptop0) and Switch1 (PC1)

| Device  | Interface | IP Address  | Subnet         | Gateway |
|---------|-----------|-------------|----------------|---------|
| Router0 | Gig0/0    | 192.168.1.1 | 192.168.1.0/24 | —       |
| Router0 | Gig0/1    | 192.168.2.1 | 192.168.2.0/24 | —       |



|         |     |              |                |             |
|---------|-----|--------------|----------------|-------------|
| PC0     | Fa0 | 192.168.1.10 | 192.168.1.0/24 | 192.168.1.1 |
| Laptop0 | Fa0 | 192.168.1.20 | 192.168.1.0/24 | 192.168.1.1 |
| PC1     | Fa0 | 192.168.2.10 | 192.168.2.0/24 | 192.168.2.1 |

### 3. Configuration

```
Router0(config)#interface GigabitEthernet0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0(config)#interface GigabitEthernet0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
Router0#write memory
```

### 4. Verification

```
PC0>ping 192.168.1.20
PC0>ping 192.168.2.10
Laptop0>ping 192.168.2.10
Router0#show ip route
```

### 5. Key Takeaways

- One router + two interfaces = two subnets, two broadcast domains.
- PC0 and Laptop0 share 192.168.1.0/24 — switch handles their traffic internally.
- Cross-subnet traffic always passes through the router — no shortcut exists at Layer 2.
- Connected routes are automatic — no static routes needed when both interfaces are up.
- Laptops in Packet Tracer can be wired or wireless — here Laptop0 uses a wired port.